Module 07 · Neurological Emergencies — Faculty Guide

PCCN Certificate Course · IAP Intensive Care Chapter

For faculty recording the five Module 07 lecture videos (V1–V5), running its live webinar and building its two simulation scenarios (7A and 7B). The format matches the Modules 01–03 guide: everything is sorted into four categories, so you know what is non-negotiable and what is optional.

Category	Meaning
Must follow	Production rules for every video.
Must cover	Clinical content each video has to teach. The module quiz is written from these points.
Better to follow	Recommended teaching style. Strongly encouraged, not checked.
Good to know	Optional pearls if time allows. Label them “beyond the exam”.

Module aim: Manage seizures, recognise raised intracranial pressure (ICP), and deliver neuro-protective nursing care, including cluster care and EVD care.

Learning objectives (from the PCCN Compendium, Section 5: Neurological Emergencies — Module 07 in the portal)

- Score conscious level accurately using AVPU and the paediatric Glasgow Coma Scale.
- Run the status epilepticus algorithm within the correct time windows.
- Apply the 10 commandments of raised ICP management and deliver cluster care.
- Care safely for a child with an external ventricular drain (EVD), including recognising shift and drift.
- Recognise and start time-critical care for CNS infection and acute encephalitis syndrome.

1. Must follow — production rules

The production rules are the same as for Modules 01–03:

1. **Length:** stay within the cap for your video (±2 min). If it runs longer, split it rather than speed up.
2. **Template:** use the PCCN slide template, with a title card showing the module number and an end card saying “Next: quiz / simulation”.
3. **Format:** 1080p, 16:9, MP4, with clear audio.
4. **Privacy:** no patient names, faces, bed numbers or IDs. Seizure or posturing videos need written consent and blurred faces.
5. **Evidence:** follow current AHA PALS, APLS and IAP guidance.
   - Every dose is per kg with the maximum dose, using generic names and a standard reference.
   - Doses must match the course’s dose table in section 5 below.
6. **No commercial content:** no logos or brand names.
7. **Language:** British spelling and SI units, speaking to the bedside nurse (“you”).
8. **Quiz alignment:** every “Must cover” point is taught clearly enough that the module quiz can be answered from the video.
9. **Upload:** YouTube Unlisted or Google Drive. Name the file PCCN_M07_V1_Title.

2. Must follow — running order of every video

Time	Segment
0:00–0:15	Title card: PCCN · Module 07 · number · title · faculty name
0:15–1:00	Bedside case hook: one child, one problem, one question
1:00–1:30	Learning objectives (at least 3)
Core	2–4 teaching chunks, each ending with a concrete nurse action; one “pause and decide” prompt
Last 1:30	Return to the case, then 3 take-home points, then a pointer to the quiz or simulation
0:10	End card

Use the **DIRECT** approach as the backbone of every case:

- **Detection:** the neurological status is assessed and named.
- **Intervention:** the immediate treatment.
- **Reassessment:** neuro observations after every intervention.
- **Effective communication:** ISBAR to the doctor, plus honest updates to the family.
- **Transport:** right place, safe transfer, airway protected.

3. Better to follow — teaching style

- Show **real bedside skills**: the GCS on a live or simulated child, pupil checks with a pen torch, recovery position, levelling an EVD to the tragus, and drawing up a buccal dose.
- Use **short, consented clips**: a focal seizure, decorticate versus decerebrate posturing, and a stridulous airway after a seizure.
- Put a **seizure clock** on screen in V2 so nurses feel the time windows.
- Focus on **what the nurse does, sees, times and escalates**, not on physician decisions.
- Provide an English subtitle file (.srt) if possible.

4. Video-by-video brief

V1 · Neurological Assessment — AVPU, Paediatric GCS & Pupils (≤12 min)

Must cover

- **AVPU:**
  - “P” (responds only to pain) or “U” means an emergency.
  - AVPU “P” corresponds roughly to GCS 8. Protect the airway and call for help.
- **Paediatric GCS:**
  - Eye, verbal and motor scores.
  - The modified verbal score for pre-verbal children (coos/babbles → irritable cry → cries to pain → moans to pain → none).
  - Always report the three components, not just the total.
- **Pupils:** size, equality and reaction to light. A unilateral fixed, dilated pupil is a herniation warning; tell the doctor immediately.
- **Posture and tone:**

    - Decorticate (abnormal flexion) and decerebrate (abnormal extension) posturing.
    - Flaccidity.
    - Neck stiffness.
- **Glucose:** check bedside glucose in every drowsy or seizing child. Treat hypoglycaemia at once using the course dose (section 5).
- **Late signs of raised ICP:**
    - Cushing's triad: hypertension, bradycardia, irregular breathing.
    - These are pre-herniation signs, not early warnings.
- **Neuro observation chart:** how often to record (every 15 minutes while unstable) and what change triggers escalation (a fall of 2 or more points in GCS, a new pupil change, or a new focal sign).

Good to know: FOUR score; the developmental milestones that change the verbal score; why sedation and paralysis invalidate the GCS.

V2 · Status Epilepticus — the Seizure Clock (≤14 min)

Must cover

- **Definition:** a seizure lasting more than 5 minutes, or repeated seizures without recovery in between. A child brought in actively seizing is treated as status epilepticus. The ILAE time points t1 and t2 are in the table.

Seizure type	Treat by (t1)	Risk of long-term harm after (t2)
Tonic–clonic	5 min	30 min
Focal, impaired awareness	10 min	more than 60 min (ILAE; the Compendium says 30 min, so align it)
Absence status	10–15 min	not known

- **0–5 min, Stabilise:**
    - Note the start time.
    - Airway: recovery position, suction, oxygen.
    - Attach monitoring: SpO_2 and heart rate.
    - Check glucose.
    - Get IV/IO access.
    - Weigh or estimate the weight.
- **5–20 min, Terminate with a benzodiazepine:**
    - Give the first dose by the fastest available route: IV/IO, or buccal or intranasal if there is no access.
    - Give one repeat dose after 5 minutes if the seizure continues.
    - **Never more than two benzodiazepine doses**, counting any dose given before arrival.
- **20–40 min, second-line drug:**
    - Levetiracetam, phenytoin/fosphenytoin, valproate or phenobarbitone, at the course doses.
    - Rate of infusion, compatibility (phenytoin with saline only), and ECG/BP monitoring.
- **More than 40 min, refractory status:**
    - Call the PICU or anaesthesia team for airway support and an infusion.
    - The nurse prepares for intubation using the SOAPME checklist.

- **Nursing during and after the seizure:**
  - Watch for respiratory depression after benzodiazepines.
  - Describe the seizure: onset, side, duration, eyes, colour.
  - Safety: nothing in the mouth, padded rails.
  - Post-ictal observations.
  - Keep the family informed.
- **Find the cause:** fever, hypoglycaemia, sodium or calcium disturbance, infection, trauma, toxins, and missed antiepileptic doses.

Good to know: non-convulsive status and when EEG is needed; ketamine and midazolam infusions in refractory status; febrile seizure versus status.

V3 · Raised ICP — the 10 Commandments (≤14 min)

Must cover

- **Physiology made simple:** the skull is a fixed box (Monro–Kellie), and CPP = MAP − ICP.
  - Hypotension harms the brain as much as a high ICP does.
  - Target CPP of about 40–50 mmHg according to age (Paediatric TBI guidelines 2019).
- **Recognition:**
  - Early signs: headache, vomiting, irritability, a falling GCS, a bulging fontanelle.
  - Late signs: a unilateral dilated pupil, abnormal posturing, Cushing's triad.
- **The 10 commandments:**

#	Commandment	Target
1	Position	head midline, head end up 30°, nothing tight around the neck
2	Cluster care	minimise stimulation
3	Sedation and analgesia	adequate
4	Glucose	below 180 mg/dL, no hypoglycaemia
5	Sodium	145–150 mmol/L
6	PaO_2	80–120 mmHg
7	$PaCO_2$	35–40 mmHg
8	Brief hyperventilation and 100% O_2	only for signs of herniation
9	Temperature and seizures	normothermia; seizures treated aggressively
10	Hyperosmolar therapy	3% saline or mannitol at the course dose

- **Nurse actions:**
  - Prepare and give hyperosmolar therapy.
  - Monitor sodium and serum osmolality (stop mannitol above 320 mOsm/kg).
  - Keep an accurate intake–output chart.
  - Check urine output after mannitol.

- Recognise and report herniation at once.

Good to know: ICP waveforms (P1/P2/P3); decompressive craniectomy nursing; suctioning technique with a raised ICP.

V4 · Caring for the Child with an External Ventricular Drain (≤10 min)

Must cover

- **What an EVD is and why it is used:** to drain CSF and measure ICP.
- **Levelling:**
  - Zero at the tragus (the level of the foramen of Monro).
  - Re-level after every change in head or bed position.
  - Height as prescribed (cmH_2O).
- **Clamping:** clamp for repositioning, transfer, coughing or suction as the unit protocol directs. **Never leave it clamped** unattended.
- **Hourly chart:**
  - CSF volume and colour: clear, blood-stained or cloudy.
  - Report over-drainage or no drainage, a change in colour, and a fall in GCS.
- **Infection control:**
  - Aseptic handling.
  - Closed system, with sampling only by trained staff.
  - Daily dressing check.
- **Recognising problems:**
  - Shift and drift of the catheter (definitions as in the Compendium).
  - Disconnection, kinking and air in the line.
  - Signs of meningitis.

Good to know: transporting a child with an EVD; weaning trials; lumbar drains.

V5 · CNS Infections & Acute Encephalitis Syndrome (≤12 min)

Must cover

- **Recognition:** fever with altered sensorium or new seizures is acute encephalitis syndrome (AES) until proven otherwise.
  - Meningitis signs: neck stiffness, bulging fontanelle, and in infants poor feeding and irritability.
  - Rash: look for non-blanching petechiae and purpura.
- **Common causes in India:**
  - Bacterial meningitis.
  - Japanese encephalitis.
  - Herpes simplex encephalitis.
  - Scrub typhus.
  - Dengue.
  - Cerebral malaria.
- **Time-critical care:**
  - ABCDE first.
  - Blood cultures, then empirical antibiotics within 1 hour of arrival (ceftriaxone ± aciclovir at the course doses).
  - Do not delay antibiotics for a lumbar puncture.

- **Lumbar puncture:** the nurse knows the contraindications (signs of raised ICP, an unstable child, coagulopathy, local infection) and positions and monitors the child during the procedure.
- **Infection control:**
  - Droplet precautions for suspected meningococcal disease for 24 hours after starting effective antibiotics.
  - Notification as per the state programme.
- **Supportive care:** seizures, fluids, sodium monitoring, fever control, glucose, and raised ICP care from V3.

Good to know: the national AES/JE programme; chemoprophylaxis for close contacts; the role of dexamethasone.

5. Course dose table — must follow in every video and simulation

Faculty must not teach doses that differ from this table. **Dr Manju Kedarnath (course director) to sign off before recording.**

Drug	Dose	Maximum	Notes
Midazolam, buccal or intranasal	0.3 mg/kg (buccal) · 0.2 mg/kg (intranasal)	10 mg	When there is no IV/IO access
Midazolam IV/IO	0.15 mg/kg	7.5 mg	Repeat once after 5 min if needed
Lorazepam IV/IO	0.1 mg/kg	4 mg	Repeat once after 5 min if needed
Diazepam IV / rectal	0.2–0.3 mg/kg IV · 0.5 mg/kg rectal	10 mg IV · 20 mg rectal	
Levetiracetam IV	40–60 mg/kg over 5–15 min	3000–4500 mg (protocol)	Second line
Phenytoin IV	20 mg/kg over 20 min	1500 mg	Saline only; ECG and BP monitoring
Fosphenytoin IV	20 mg PE/kg	1500 mg PE	
Sodium valproate IV	40 mg/kg over 5–10 min	3000 mg	Avoid under 2 years and in suspected metabolic or liver disease
Phenobarbitone IV	20 mg/kg over 20 min	1000 mg	Watch for respiratory depression and hypotension
3% sodium chloride	3–5 mL/kg over 10–20 min	—	Raised ICP / herniation
Mannitol 20%	0.5–1 g/kg (2.5–5 mL/kg) over 20 min	—	Stop if osmolality is above 320 mOsm/kg
10% dextrose (hypoglycaemia)	**Course to choose one:** 2 mL/kg (APLS, used in Modules 01–03) **or** 10 mL/kg (Compendium Section 5)	—	Recheck glucose in 15 min
Ceftriaxone IV	100 mg/kg/day in 1–2 doses (meningitis)	4 g/day	Within 1 hour
Aciclovir IV	20 mg/kg 8-hourly (under 3 months) · 10–15 mg/kg 8-hourly (older)	per protocol	Suspected HSV encephalitis

Flag for the course director: the Compendium's 10% dextrose dose (10 mL/kg) differs from the dose already used in Modules 01–03 (2 mL/kg, APLS). Choose one dose for the whole course before scripts are written.

6. Live webinar — “Neuro Night Shift: GCS Calibration & the Seizure Clock” (90 min · Step 2 attendance)

The webinar follows the running order and rules in the **PCCN Live Webinars Faculty Guide**, with a moderator, lead faculty, breakout facilitators and a tech host. Delegates need 70% attendance; anyone who misses it watches the recorded makeup.

Must cover

- **GCS calibration:** three consented clips (infant, toddler, school-age child). Every delegate scores E, V and M in a poll, then the faculty reveal the agreed scores and discuss the errors.
- **Seizure clock, run live:** a 3-year-old is still seizing after diazepam given by the GP. A visible timer runs, with a decision at 0–5, 5–10, 20 and 40 minutes, and the second-line dose calculated live.
- **Breakout drill:** a “10 commandments” cluster-care plan for the first 4 hours of a child with meningitis and raised ICP, including the triggers for calling the doctor.
- **EVD demonstration:** levelling to the tragus, clamping, the hourly chart, shift and drift.
- **Communication:** an ISBAR role-play for a falling GCS, and how to talk to parents during a prolonged seizure.

Materials: GCS clips, a seizure-clock slide with a timer, a case card with an answer key, an EVD set or photo, and the dose sheet.

7. Simulations — 2 scenarios

Must follow, for every scenario:

- Use the PCCN virtual-sim framework: case, then video, then Impression, then Intervention, then response, then Reassessment, then Communication with the parents, then Team communication, then Transfer, then outcome.
- Every step is “choose all that apply”, graded Correct, Acceptable or Unsafe.
- Every unsafe choice shows its consequence on the monitor.
- Faculty provide a storyboard: one row per decision, listing the options, the grade of each, the reason, and how the vitals change.

Scenario	Case	Decision points (DIRECT)
7A · The Seizure Clock	3-year-old, 14 kg, brought in actively seizing for 15 min at home with fever; one dose of rectal diazepam given by the GP	**D** the child is in status epilepticus (unstable, life-threatening, neurological) → **I** airway, O_2, glucose and IV access, then IV midazolam or lorazepam as the *second and last* benzodiazepine → **R** still seizing at 10 min, so prepare levetiracetam; watch for apnoea → **EC** a frightened mother; time-stamped ISBAR to the doctor → **T** PICU after seizure control, airway protected
7B · The Unequal Pupil	7-year-old, 22 kg, day 2 of meningitis; GCS falls from 13 to 8, with BP 130/80, HR 62 and irregular breathing	**D** raised ICP with impending herniation (Cushing’s triad) → **I** head up 30° midline, O_2, call for help, prepare 3% saline or mannitol, no LP → **R** pupils, GCS and sodium after hyperosmolar therapy → **EC** honest update to the family; ISBAR to the PICU → **T** PICU for intubation and CT, with safe transfer

Better to follow: use a real neuro observation chart and a drug-calculation step with a dose check by a second nurse.

8. Hands-on workshop links

Station	Content
Neuro assessment OSCE	AVPU and paediatric GCS on a simulated infant and child, pupils, and neuro chart entry
Seizure management drill	Seizure clock in real time: recovery position, suction, buccal midazolam draw-up, second-line drug calculation
EVD care	Levelling to the tragus, clamping and unclamping, hourly chart, sterile sampling (demonstration)
Cluster-care planning	Group activity: plan 24 hours of care for a child with raised ICP

9. Pre-upload checklist (faculty)

- ☐ Within the length cap
- ☐ Title card and end card
- ☐ Case hook and 3 objectives
- ☐ Every “Must cover” point taught
- ☐ Doses exactly as in the course dose table, per kg with the maximum
- ☐ Recap with 3 take-home points
- ☐ No patient identifiers; seizure clips consented and blurred
- ☐ No logos or brands
- ☐ 1080p, clear audio
- ☐ File named PCCN_M07_V1_Title

10. Assignment and timeline

Task	Faculty	Script due	Record by	Review	Live
V1 Neurological Assessment					
V2 Status Epilepticus					
V3 Raised ICP					
V4 EVD Care					
V5 CNS Infections & AES					
Sim 7A The Seizure Clock					
Sim 7B The Unequal Pupil					
Live webinar (lead + moderator + facilitators)					

References

- American Heart Association. *Pediatric Advanced Life Support Provider Manual*, 2020.
- Advanced Life Support Group. *Advanced Paediatric Life Support*, 7th ed. Wiley, 2023.
- Trinka E, et al. A definition and classification of status epilepticus — ILAE Task Force. *Epilepsia* 2015;56:1515–23.
- Glauser T, et al. Evidence-based guideline: treatment of convulsive status epilepticus in children and adults. *Epilepsy Curr* 2016;16:48–61.

- Kapur J, et al. Randomized trial of three anticonvulsant medications for status epilepticus (ESETT). *N Engl J Med* 2019;381:2103–13.
- Kochanek PM, et al. Guidelines for the management of pediatric severe traumatic brain injury, 3rd ed. *Pediatr Crit Care Med* 2019;20:S1–82.
- Indian Academy of Pediatrics. *Standard Treatment Guidelines* — Status Epilepticus; Acute Encephalitis Syndrome.
- PEERS Manual 2025 — Management of Neurological Emergencies; BPCCN Guide — Raised ICP, CPP Monitoring, EVD Competencies.
- PCCN Course Compendium, Section 5: Neurological Emergencies.